Can a Machine Read Your Emotions?

Facial Expression Recognition with ResNet18 & CLIP on FER-2013

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The Experiment

"What am I feeling right now?"

< 1 sec

Time for your brain
to read an emotion

Automatic

No conscious effort —
it just happens

Our question:

Can a machine
do the same?

Why Is This Hard?

35’887

Total Images

web-scraped from the internet

48 × 48 px

Image Size

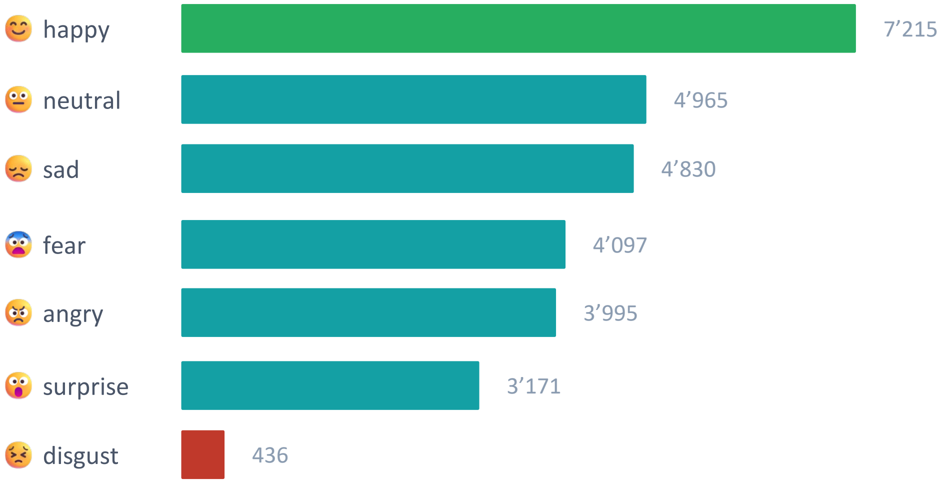
grayscale — very low resolution

7 Classes

Emotion Labels

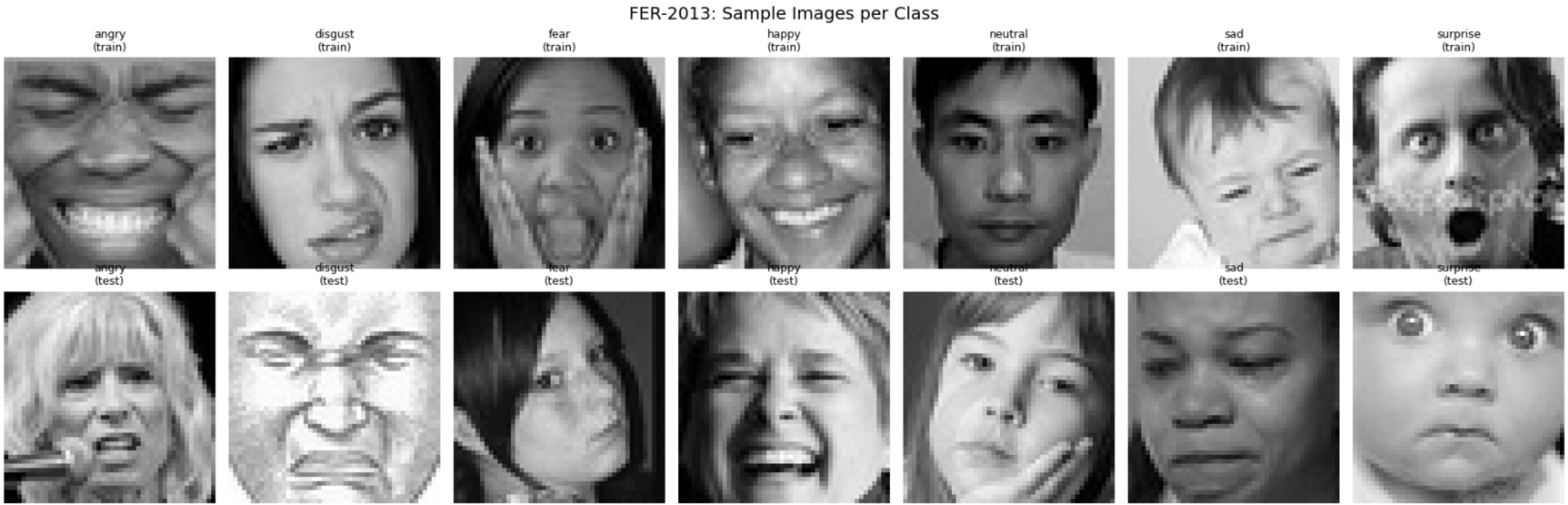
angry, disgust, fear, happy, neutral, sad, surprise

Class Imbalance Challenge



The Problem

"Can you tell fear from surprise? — Even humans only reach ~65% on the same images. The bar is lower than you'd think."



Two Ways to Teach a Machine Emotions

Approach 1: ResNet18 CNN

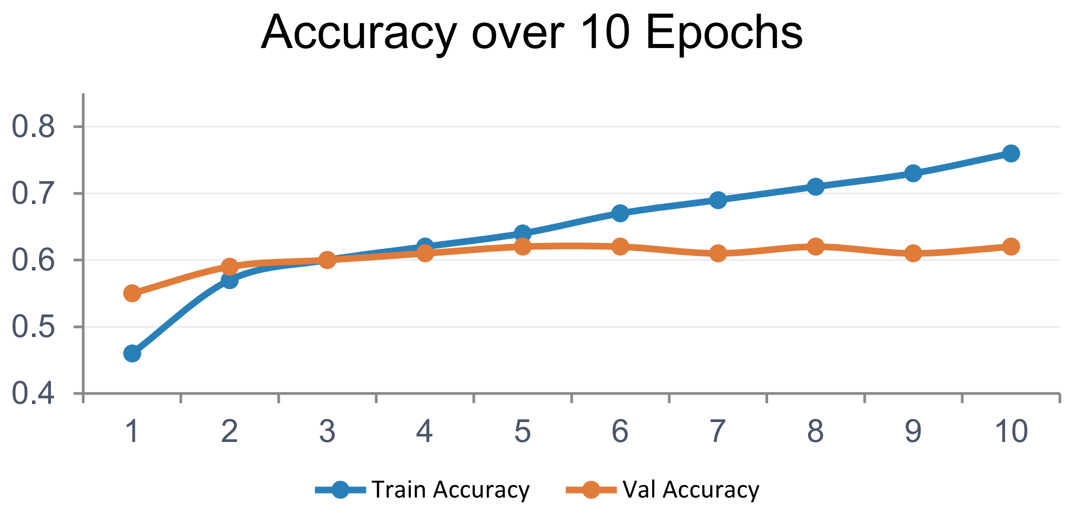
- Pre-trained on ImageNet (1.2M images)
- Frozen backbone + fine-tuned layer4
- Learns from 28,000 labelled examples
- Trained 20 epochs total
- Best Val Accuracy: ~61%
- Feature maps reveal what it "sees"

Approach 2: CLIP Zero-Shot

- Trained on 400M image-text pairs
- No task-specific training on FER-2013
- "Tell the model what to look for" — in text
- Cosine similarity: image vs. text prompt
- Zero-shot accuracy: 40.5%
- Strong baseline — no labels needed

Training & Results

~61%	~62%	20	1e-4
Val Accuracy	**Test Accuracy**	**Epochs**	**Learning Rate**
Best checkpoint (Epoch ~5)	Held-out test set	10 + 10 extended training	Adam + ReduceLROnPlateau

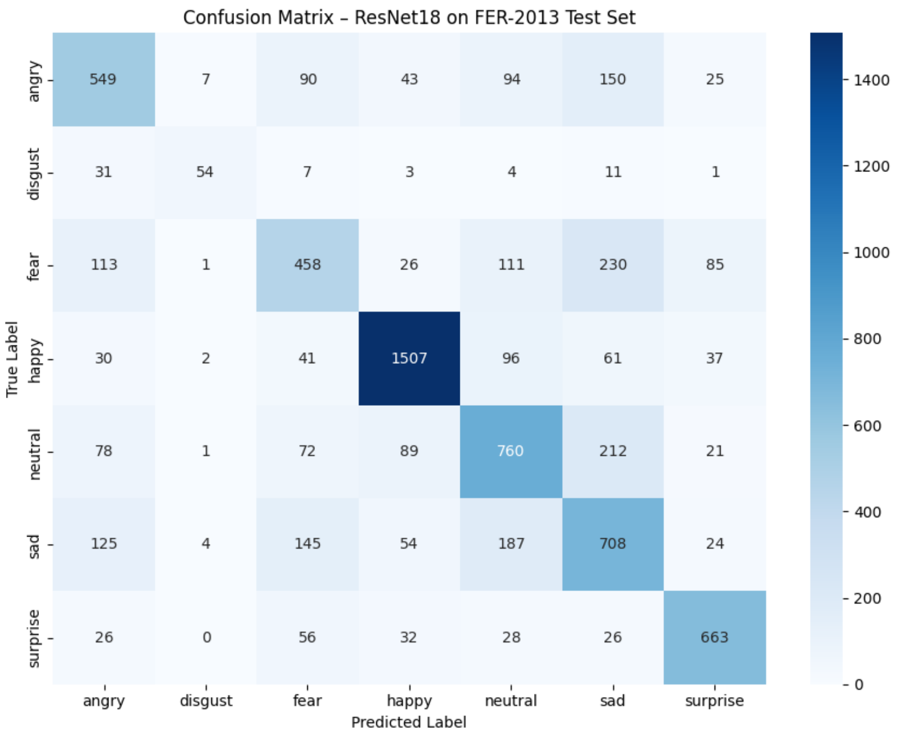


Key Observations

- Val accuracy stabilised at ~61%
- Dropout(0.5) reduced overfitting
- Frozen backbone → fast training
- layer4 fine-tuning critical (+17% vs frozen-only)
- Train/Val gap → Overfitting after Epoch 5
- FER-2013 benchmark: ~75% SOTA

Evaluation: Confusion Matrix

Best recognised emotions vs. most confused pairs



Top Classes

Happy 85%, Surprise 80%, Neutral 62%

Lowest-performing Classes

Fear 44%, Disgust 49%

Most Common Confusions

Fear <-> Sad (230)
Sad <-> Neutral (187)
Angry <-> Sad (150)

What Does the Network "See"?

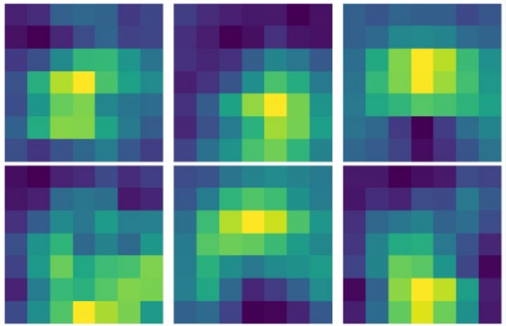
layer1 — Early Features

Detects low-level patterns:

- Edges and corners
- Colour gradients
- Texture variations

Similar across ALL emotions → generic ImageNet features → can be frozen

layer4 — Deep Features



Detects high-level patterns:

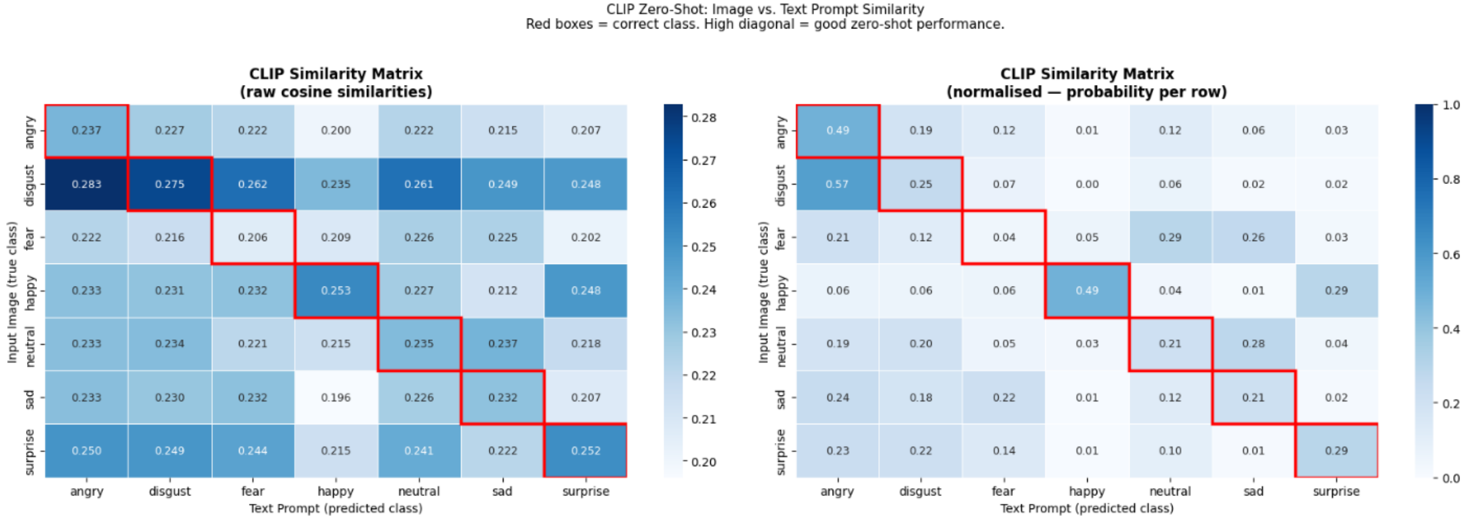
- Facial expressions & muscle groups
- Eye/mouth configurations
- Abstract emotion-relevant features

Sparse, task-specific → DIFFERENT patterns per emotion → fine-tune this layer!

CLIP: What If You Just Describe the Emotion?

CLIP classifies without any labelled training data by comparing image embeddings to text prompt embeddings via cosine similarity.

Image Input — CLIP Image Encoder — Image Embedding — **Cosine Similarity** — Predicted Emotion



Live Demo on Our Own Photos

1 Make a face

Pick an emotion
— exaggerate it!
Upload a photo live.

2 Face detection

OpenCV Haar Cascade detects
face region
+ 10% padding.

3 Preprocessing

Resize 224×224
Grayscale→RGB
ImageNet normalisation.

4 Prediction

ResNet18 outputs
confidence score
for all 7 classes.

Limitations

- Domain gap: 48×48 grayscale ≠ real-world colour photos
- Disgust class: only 436 training images → poor recall
- No face alignment / pose normalisation

Conclusion

Task:	7-class emotion recognition on FER-2013 with ResNet18
Result:	62% test accuracy — solid baseline, not far from human level (~65%)
Key finding:	Fine-tuning layer4 gave +17% over frozen backbone (36% → 53% → 62%)
CLIP:	CLIP zero-shot reaches 40.5% — surprisingly strong without any FER training
Limitation:	Domain gap: model struggles on real-world colour photos

Questions?